Project Plan

WrenchIT

NXT Technologies

Industry Partner	
Primary Instructor	Prof. Laily Ajellu
Team Member	Tyson Ward-Dicks
Team Member	Hossein Khanzadeh
Team Member	Henrique Custodio
Team Member	Hamzah Hafez
Team Member	Frederich Tan

Document Revision History

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1. Executive Summary

The following describes the project to be executed.

Objective	Wrench It is a digital platform for connecting vehicle owners with local mechanics and autobodyshops. Providing search listings, map-based locations, review and price comparison to improve transparency. The platform enables users to make informed decisions by comparing services, prices, and customer feedback in one convenient interface. Wrench It helps users quickly find trusted local mechanics, making vehicle maintenance simpler and more reliable.
Corporate Goals Addressed	Promote innovation in untapped markets
Planned Start Date	September 2025
Planned End Date	December 2025

2. Project Approvers, Reviews and Distribution List

Approvers, reviewers and distribution list

Project Role	Name	E-mail	Date
Developer Lead	Henrique Custodio	henrique.custodio@georgebrown.ca	10/23/2025
Developer - Data Analytics	Tyson Ward-Dicks	tyson.ward-dicks@georgebrown.ca	10/23/2025
Developer - Backend	Hamza Hafez	hamzah.hafez@georgebrown.ca	10/23/2025
Developer - UI Design	Hossein Khanzadeh	hossein.khanzadeh@georgebrown.ca	10/23/2025
Database design	Fredrich Tan	fredrich.tan@georgebrown.ca	10/23/2025

3. Scope

Define the sum total of all of its products and their requirements or features.

In Scope	Out of Scope
Use a map integration (API) to show locations	Creating a payments system
Add a search feature	Advanced data and AI
Add review features for customers	In-App Bookings and appointments
Display simple shop info	Advanced UI/UX (multi-language support, and accessibility features…)
User Authentication System	Online payments
Basic user profile editing	Real time chat with mechanics

4. Deliverables

This project will deliver the following.

Deliverable	Description	Mini Tasks
Search and Filter Feature	A functional system that will allow users to search for nearby mechanic shops and view their details.	1. Implement search bar and filtering options 2. Connect search functionality to the backend database 3. Display search results dynamically with shop details
Shop Information and Map Integration	Display mechanic shop details such as name, address, contact info, operating hours, and show shop locations visually using Google Maps API.	1. Design shop info layout and pages 2. Integrate Google Maps API for location visualization 3. Link map pins to detailed shop info pages
User Authentication and Profiles	Allow users to simply register, login, and manage their accounts.	1. Implement user signup and login system 2. Enable profile editing (name, email, password) 3. Securely store user data in the backend database

Customer Review and Rating System	Allow users to rate and leave reviews and feedback for different shops. Anyone will be able to view the ratings for the shops.	1. Create review submission form with rating options 2. Connect reviews to shop database entries 3. Display reviews and average ratings on shop pages
Backend Database Management	Maintain a database to store all relevant data for the app, including user information, mechanic shop details, services, and reviews.	1. Design database schema for users, shops, services, and reviews 2. Implement CRUD operations 3. Maintain accurate and secure data through careful validation and database organization.

5. Assumptions

This project makes the following assumptions;

- Users will have access to a stable internet connection
- Mechanic shops information will be available
- Team members will have access to various collaboration tools
- API's will function properly
- Users will use modern smartphones that will be compatible with the app's requirements
- The databases will be reliable and maintain data integrity throughout usage.
- Users will provide honest reviews for mechanic shops
- Users will have enough basic digital literacy to be able to navigate the app
- Each team member will contribute equally
- All team members will have enough technical skill to complete their assigned tasks

6. Dependencies

The following are the internal and external dependencies that will have to be acknowledged and addressed;

- Team members must complete their tasks on time
- Proper communication and coordination between team members

- The version control must be used properly between members to avoid merging and other conflicts
- Availability and reliability of third party API's
- Reliable internet connections for users
- Mechanic shops data provided to third parties being accurate
- Availability of a stable hosting environment or server for the application
- Access to development tools, libraries, and frameworks required for building the app
- Users enabling location services for accurate map-based searches
- Consistent testing environment to ensure features work as expected

7. Risk Management

Potential Risk	Severity (H/M/L)	Likelihood (H/M/L)	Management Strategy
API failure	H	M	Test alternative API's
Inaccurate Mechanic shop data	M	H	Validate data before adding to the database
Changes in project requirements	L	M	Make sure to document all changes and get approval from all team members
Team member Drops out	H	L	Reassign tasks among remaining members
Minor version control conflict on GitHub	L	M	Enforce frequent commits, pull requests and make sure to properly merge.
Team coordination and communication delays	L	H	Use group chats to keep everyone updated on the project process. Assign members to weekly goals.
Browser compatibility issues	M	M	Test the app across multiple modern browsers and devices during development.

8. Communication

Reporting

The following reports will be produced;

Report	Audience	Frequency
Progress Report	Instructor	Per Sprint
Sprint Retrospective	Team	End of each sprint
Final Report	Prof.	End of semester

Meetings

The following meetings/communication will be established;

Meeting	Purpose	Attendees	Frequency
Sprint Planning	Defining weekly goals, tasks & dependencies	All team members	Weekly
Progress Checks	Review completed tasks and blockers	All team members	Weekly
Code Review	Ensuring code quality & Version	Dev Lead + UI/Backend devs	Bi-weekly
Instructor Feedback	Getting academic and project guidance	Instructor + Team	As scheduled

9. Task Listing (WBS- Work Breakdown Structure)

The following resource proposal template summarizes the resource hours committed to this project, upon final approval of this document.

Reference	Tasks	Duration	Dependency
A	Project setup ,Github repo & Database	1 Week	None
B	Backend api & maps	Weeks 2-4	A

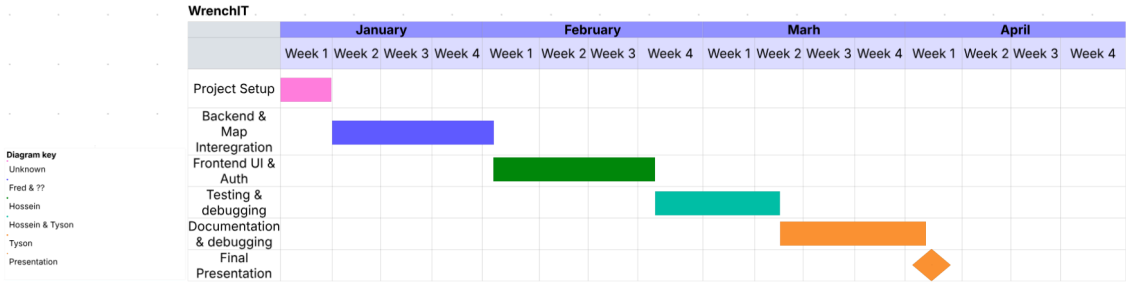
C	Frontend design	Weeks 4-6	A
D	Testing/Debugging & Data validation	Weeks 7-9	B,C
E	Documentation & prototype presentation	Weeks 10-12	D
F	Submission & Final Presentation	Week 13	E

10. Gantt Chart

Create a detailed Gantt Chart from your Task Listing(Use any software tool and paste the image or upload as a separate file that can be opened as pdf/doc/xls)

Below is an example:

Task	Period								Completed
	Dates	Dates	Dates	Dates	Dates	Dates	Dates	Dates	
Project Set-Up	Jan6							Jan 12	Planned
Backend & Map Integration	Jan 13							Feb 2	Planned
Frontend UI & Auth	Feb 3							Feb 16	Planned
Testing & Debugging	Feb 17							Mar 9	Planned
Documentation & Debugging	Mar 10							Mar 30	Planned
Final Presentation	Apr 1							Apr 6	Planned



11. Milestones

Major Activity or Milestone	Estimated Milestone Target date	Owner/Reviewer Team Members
Project Set-Up Complete	Jan 12	Henrique
Business Logic + API ready	Feb 2	Hamza
Map and Search Integration Complete	Feb 9	Fredrich Tan
Frontend Prototype Functional	Feb 23	Hossein
Review System Integrated	Mar 2	Hossein & Tyson
Quality Assurance & Data Validation Complete	Mar 9	Tyson
Final Presentation & Report	Apr 6	All members

12. RAM – Responsibility Assignment Matrix

Project Team Responsibilities

Project Name: Wrench*IT*
Project Manager: NXT Technologies

Task	**Henrique**	**Hossein**	**Fredrich**	**Tyson**	**Hamzah**
Project Set up	P		P		P
Backend & map integration	P			S	P
Frontend UI & Auth		P	S		
Testing & Debugging	P	S	P	P	S
Documentation & Debugging			S	P	P
Final Presentation	P	P	P	P	P

13. Approval

The signatures below indicate their approval of the contents of this document.

Project Role	Name	Signature	Date
Developer Lead	Henrique Custodio	*Henrique Custodio*	12-02-2026
Developer	Hamzah Hafez	*Hamzah Hafez*	12-02-2026

Developer - UI Design	Hossein Khanzadeh	Hossien Khanzadeh	12-02-2026
Developer	Tyson Ward-Dicks	Tyson Ward-Dicks	12-02-2026
Developer	Fredrich Tan	Frederich Tan	12-02-2026
Instructor	Ms. Laily Ajellu		